AP-001

Site Alarm Philosophy and Rationalisation Standard

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1 Purpose

This standard defines how alarms are prioritised, rationalised and assessed for performance across the site. It is aligned to ISA-18.2 and applies to all alarms presented to a human operator.

3 Priority Assignment

Priority is assigned from the consequence of inaction and the time available to respond. Priority is a property of the alarm, not of the asset; a low-consequence alarm on a critical asset does not inherit that criticality.

Consequence	Under 5 min	5 to 30 min	Over 30 min
Safety or environmental	P1 Critical	P1 Critical	P2 High
Major production loss	P1 Critical	P2 High	P3 Medium
Equipment damage	P2 High	P2 High	P3 Medium
Minor or recoverable	P3 Medium	P3 Medium	P4 Low

Table 3-1 — Priority matrix

3.2 Composite Priority Score

Where a numeric score is required for ranking, the composite score is computed from alarm severity, asset criticality and recurrence, weighted 0.35, 0.40 and 0.25 respectively, and normalised to a 0 to 100 scale. The score supports triage across alarms; it does not override the priority assigned under Table 3-1.

5 Performance Targets

Metric	Target	Maximum acceptable
Alarms per operator per hour	6	12
Alarms in 10-minute flood window	Under 10	10
Priority 1 share of total	Under 5 percent	5 percent
Standing alarms at shift change	Under 5	10
Mean acknowledgement delay	Under 5 min	15 min

Table 5-1 — Alarm system performance targets

5.2 Reportable Deviations

A mean acknowledgement delay above 15 minutes sustained over any 30-day period is reportable to the site alarm management team, irrespective of alarm priority.

6 Rationalisation

6.1 Candidate Criteria

An alarm becomes a rationalisation candidate where any of the following holds over a 30-day assessment window:

- It recurs more than five times without a corresponding operator action being recorded, indicating a nuisance or chattering alarm.
- It remains active for more than 180 minutes without acknowledgement, indicating a standing or stale alarm.
- It is consistently accompanied by another alarm with which it shares a cause, indicating a redundant or consequential alarm.
- No documented operator response exists for it.

6.2 Setpoint Changes

Alarm setpoints may only be changed through the rationalisation process. Changing a setpoint to silence a recurring alarm without addressing its cause is prohibited and is treated as a bypass.

7 Flood Definition

An alarm flood exists where more than 10 alarms are presented to a single operator within any rolling 10-minute window. Flood periods are excluded from acknowledgement delay statistics but are counted separately as a system performance failure.